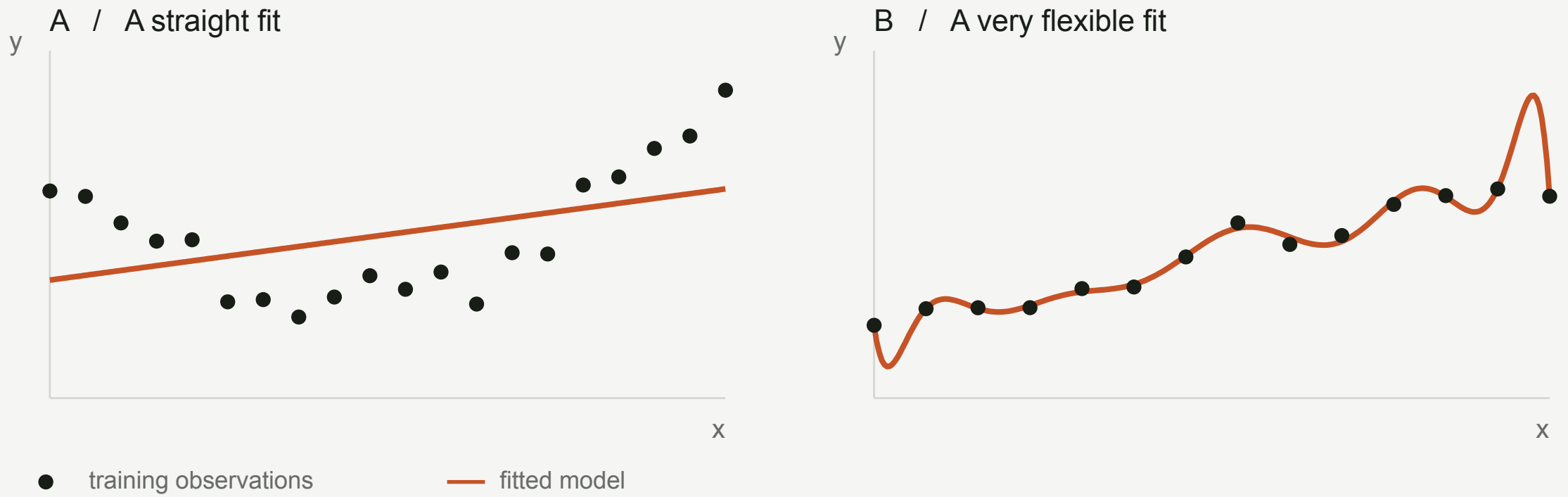
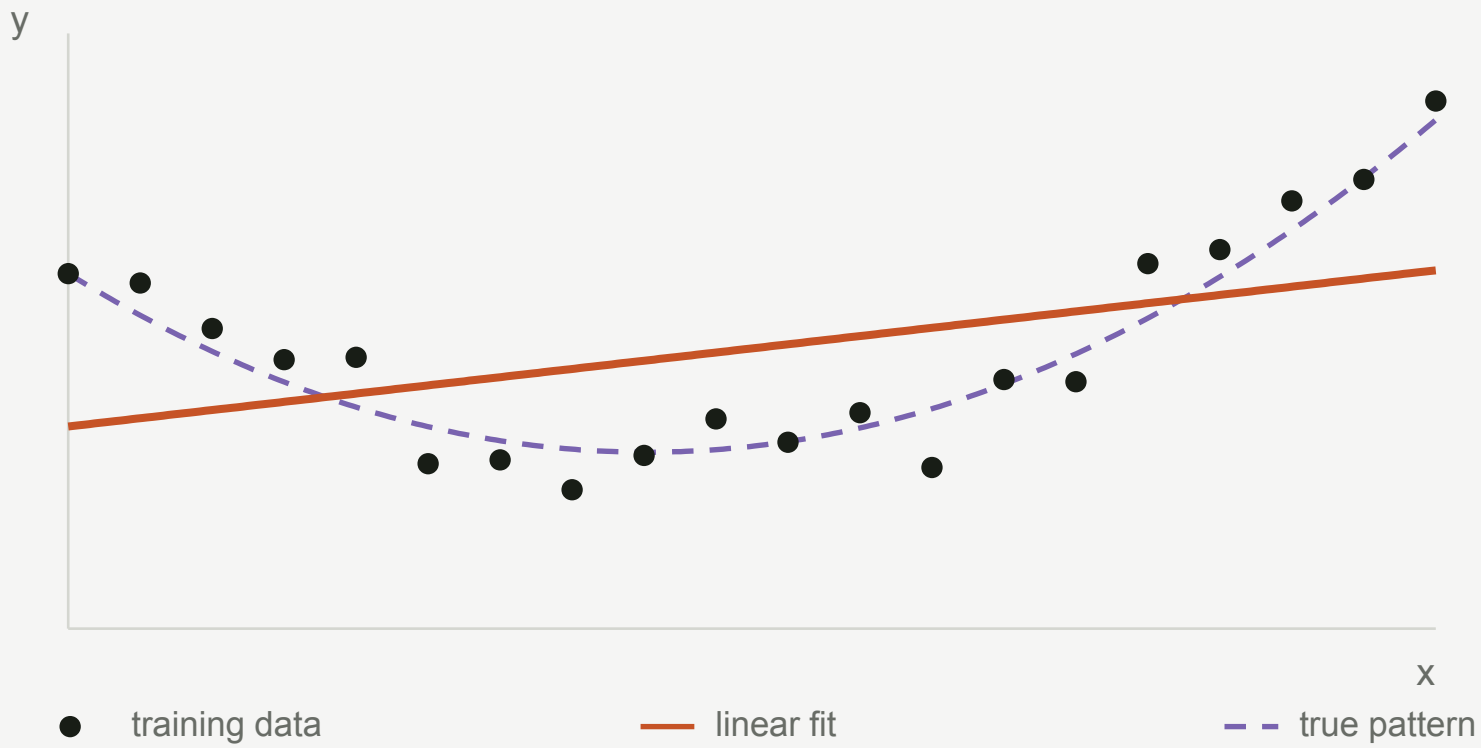
Which model would you trust?



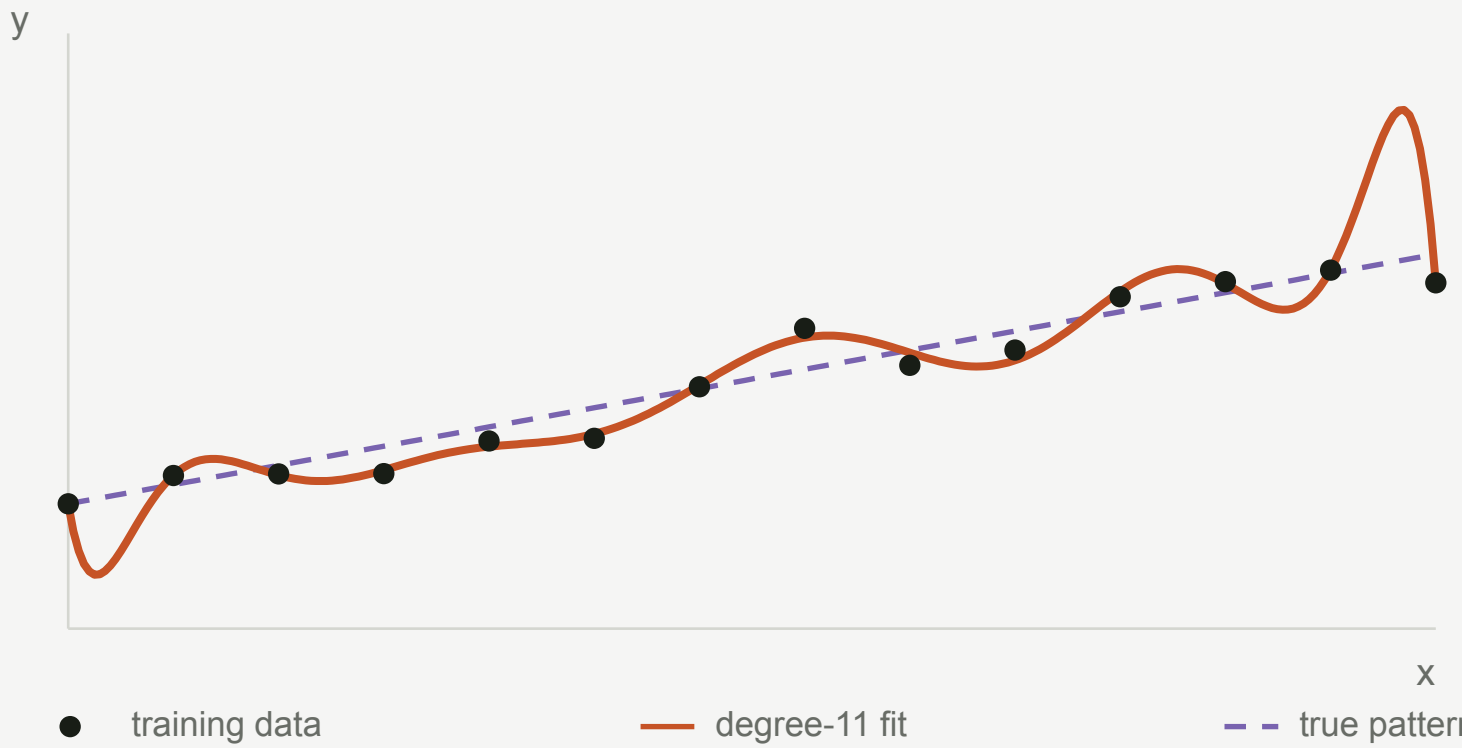
Too simple to learn the pattern.



High bias

A straight line cannot learn this curve.

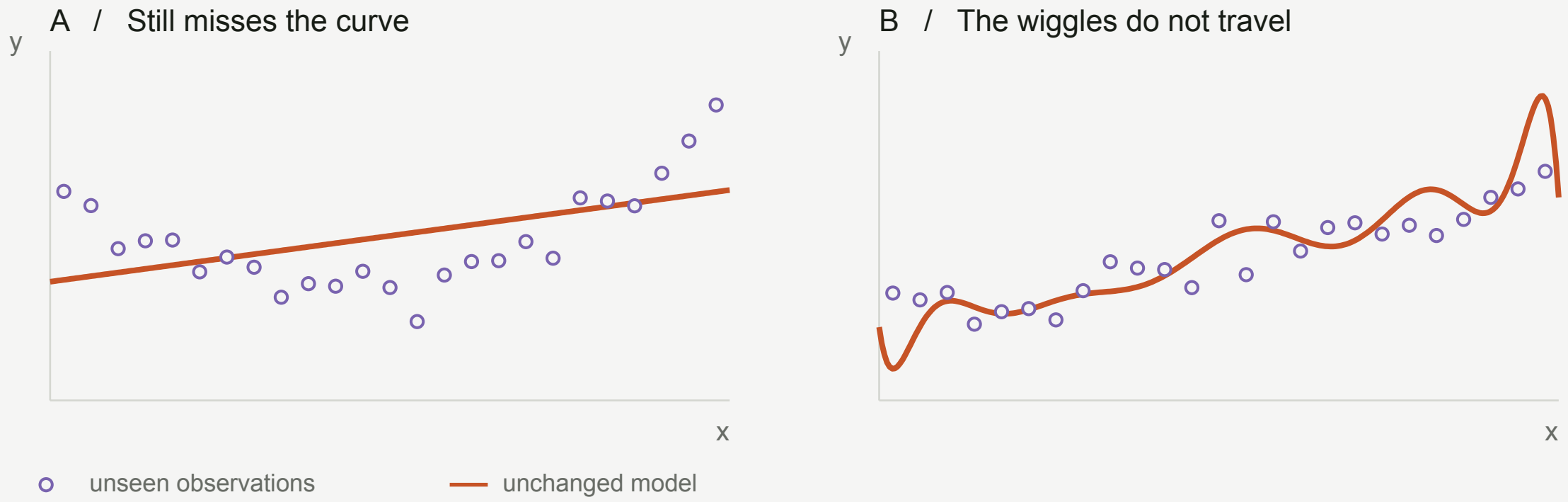
Flexible enough to learn the noise.



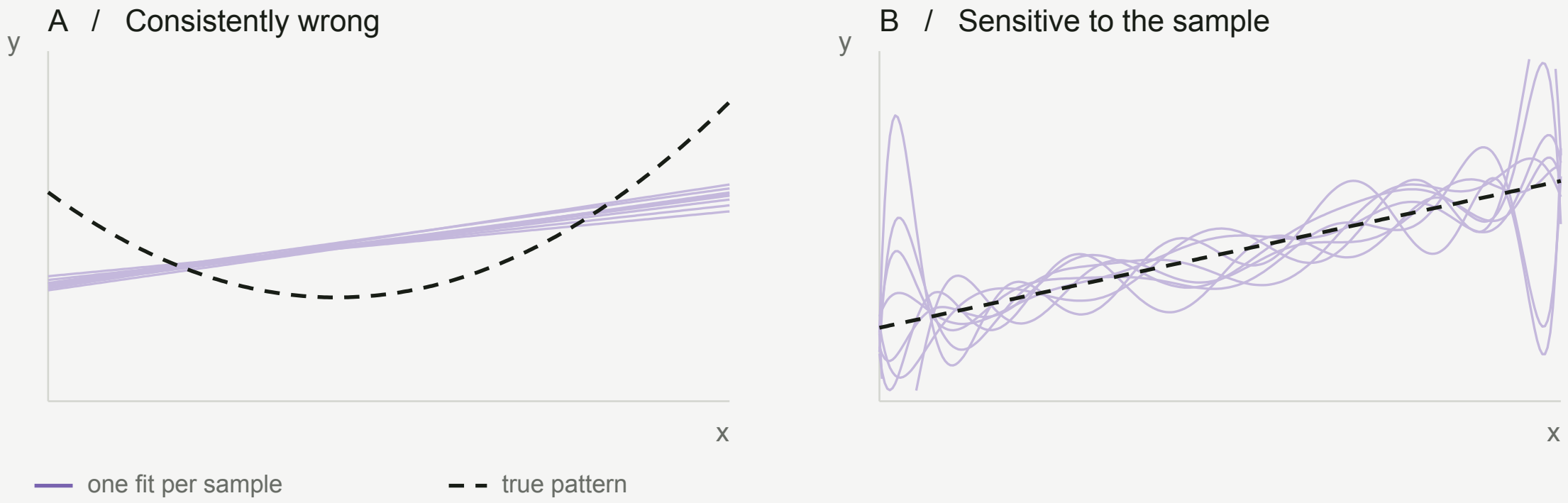
High variance

Small changes in data.
Big changes in the fit.

New data does not repeat the noise.



Change the sample. Fit again.



A closer fit is not always a better model.



Choose using validation data.

Schematic, not measured

The smallest training loss can lose.

Fit	Model degree	Training MSE ↓	Testing MSE ↓
Underfitting	1	0.251	0.183
Good fit	**2**	**0.021**	**0.027**
Overfitting	12	0.008	0.043

Learn the pattern.
Not the noise.

BIAS

Too rigid to capture the relationship.

VARIANCE

Too sensitive to the training sample.

GENERALIZATION

How well it predicts new observations.

Tune on validation. Report on the untouched test set.